บุญญารัศมิ์ 663020036-9

Hw 4

Age

<30

1	N
2	N
8	N
9	Y
11	Y

31-40

3	Y
7	Y
12	Y
13	Y

>40

4	Y
5	Y
6	N
10	Y
14	N

$$Info(D) = I(9,5)$$
$$= \left(-\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right)\right)$$
$$= 0.940$$

$$Info_{age}(D) = \frac{5}{14}I(2,3) + \frac{4}{14}I(4,0) + \frac{5}{14}I(3,2)$$
$$= \frac{5}{14}\left(-\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)\right) + \frac{4}{14}\left(-\frac{4}{4}\log_2\left(\frac{4}{4}\right) - \frac{0}{4}\log_2\left(\frac{0}{4}\right)\right) + \frac{5}{14}\left(-\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)\right)$$
$$= 0.347 + 0 + 0.347$$
$$= 0.694$$

$$Gain(age) = 0.940 - 0.694$$
$$= 0.246 \quad \#$$

income

low

5	Y
6	N
7	Y
9	Y

medium

4	Y
8	N
10	Y
11	Y
12	Y
14	N

High

1	N
2	N
3	Y
13	Y

$$Info_{inc}(D) = \frac{4}{14}I(3,1) + \frac{6}{14}I(4,2) + \frac{4}{14}I(2,2)$$
$$= \frac{4}{14}\left(-\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right)\right) + \frac{6}{14}\left(-\frac{4}{6}\log_2\left(\frac{4}{6}\right) - \frac{2}{6}\log_2\left(\frac{2}{6}\right)\right) + \frac{4}{14}\left(-\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)\right)$$
$$= 0.231 + 0.394 + 0.286$$
$$= 0.911$$

$$Gain(income) = 0.940 - 0.911$$
$$= 0.029 \quad \#$$

	age	income	student	credit_rating	buys_computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31…40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31…40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31…40	medium	no	excellent	yes
13	31…40	high	yes	fair	yes
14	>40	medium	no	excellent	no

student

	N
1	N
2	N
3	Y
4	Y
8	N
12	Y
14	N

	Y
5	Y
6	N
7	Y
9	Y
10	Y
11	Y
13	Y

$info_{student}(D) = \frac{7}{14}(3,4) + \frac{7}{14}(6,1)$

$= \frac{7}{14}\left(-\frac{3}{7}\log_2\left(\frac{3}{7}\right) - \frac{4}{7}\log_2\left(\frac{4}{7}\right)\right) + \frac{7}{14}\left(-\frac{6}{7}\log_2\left(\frac{6}{7}\right) - \frac{1}{7}\log_2\left(\frac{1}{7}\right)\right)$

$= 0.493 + 0.296$

$= 0.789$

$Gain_{stud} = 0.940 - 0.789$

$= 0.151$ ※

Credit_rating

	fair
1	N
3	Y
4	Y
5	Y
8	N
9	Y
10	Y
13	Y

	excellent
2	N
6	N
7	Y
11	Y
12	Y
14	N

$info_{credit}(D) = \frac{8}{14} I(6,2) + \frac{6}{14} I(3,3)$

$= \frac{8}{14}\left(-\frac{6}{8}\log_2\left(\frac{6}{8}\right) - \frac{2}{8}\log_2\left(\frac{2}{8}\right)\right) + \frac{6}{14}\left(-\frac{3}{6}\log_2\left(\frac{3}{6}\right) - \frac{3}{6}\log_2\left(\frac{3}{6}\right)\right)$

$= 0.464 + 0.429$

$= 0.893$

$Gain_{credit} = 0.940 - 0.893$

$= 0.047$ ※

∴ จาก Gain ทั้งหมด พบว่า Gain age มีค่า = 0.246 เป็นค่ามากที่สุด จึงแบ่งแยกข้อมูลได้ดีกว่า Gain อื่น ※